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[22CSE153]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER -2023

RELATIONAL DATABASE MANAGEMENT SYSTEMS

Time: 3 Hours Maximum **Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 X 2=10**		
1.	**a.**	List the difference between Flat model and Hierarchical model.	L1	CO1	**2**
	b.	Identify the types of relational algebraic operations.	L1	CO2	**2**
	c.	Write a query to copy the structure of table from another table without the data in the existing table.	L5	CO3	**2**
	d.	Define the force view.	L1	CO4	**2**
	e.	Write a note on ACID properties.	L1	CO5	**2**
		PART-B	**6 X 5=30**		
2.	**a.**	Elaborate the characteristics of a database.	L2	CO1	**5**
		OR			
	b.	Illustrate the types of attributes with an example.	L1	CO1	**5**
3.	**a.**	Write a relational algebraic query to find the names of sailors who have reserved in boat and write a note on select operation.	L5	CO2	**5**
		OR			
	b.	Consider the following relational DB of a college. Student (Rno , name , address) Find the names of the students who live in Chennai. Delete the names of the students whose address is lucknow.	L6	CO2	**5**
4.	**a.**	Write the difference between VARCHAR and CHAR with example.	L2	CO3	**5**
		OR			
	b.	Write a short notes on aggregate functions in SQL with example.	L1	CO3	**5**
5.	**a.**	Classify the types of views in SQL with the help of example queries.	L1	CO4	**5**
		OR			
	b.	Write a PLsql function to reverse a number.	L1	CO4	**5**
6.	**a.**	Elaborate the method of two-phase locking.	L3	CO5	**5**
		OR			
	b.	Write a note on functional dependency and multivalued dependency with example.	L3	CO5	**5**
7.	**a.**	**i.** Write a query to display the employee details where name starts with 're'. **ii.** Write a query to display employee details in which salary contain number 50 in between.	L5	CO3	**5**
		OR			

	b.	Write a PLsql program to display the names of all countries.	L5	CO4	5
		PART-C	**6 X 10=60**		
8.	**a.**	Narrate the three Schema Architecture with a neat diagram.	L3	CO1	**10**
		OR			
	b.	Describe the classification of database languages with examples.	L2	CO1	**10**
9.	**a.**	Suppose that you are designing a schema to record information about reality shows on TV. Your database needs to record the following information: _ For each reality show, its name, genre, basic info and participant's name. Any reality show has at least two or more participants. _ For each producer, the company name, company country. A show is produced by exactly one producer. And one producer produces exactly one show. _ For each television, its name, start year, head office. A television may broadcast multiple shows. Each show is broadcasted by exactly one television. _ For each user, his/her username, password, and age. A user may rate multiple shows, and a show may be rated by multiple users. Each rating has a score of 0 to 10. Draw an entity relationship diagram for this database.	L5	CO1	**10**
		OR			
	b.	Narrate the database architecture with neat diagrams.	L2	CO1	**10**
10.	**a.**	How relational calculus is different from relational algebra? Explain.	L3	CO2	**10**
		OR			
	b.	Explain the operations of relational algebra with examples.	L2	CO2	**10**
11.	**a.**	Brief the concept of primary key and foreign key with examples. Write the data dictionary query regarding that constraints.	L1	CO3	**10**
		OR			
	b.	Create a table called employee and perform all DML operations in that table.	L5	CO3	**10**
12.	**a.**	Write a code in PL/SQL to implement a trigger that maintains a transaction history log whenever a row is deleted from a table.	L5	CO4	**10**
		OR			
	b.	Write a program in PL/SQL to Fetch multiple records with the uses of nested cursor.	L5	CO4	**10**
13.	**a.**	Given a relation R(A, B, C, D) and Functional Dependency set FD = { AB → CD, B → C }, determine whether the given R is in 2NF? If not convert it into 2 NF.	L6	CO5	**10**
		OR			
	b.	Describe 2NF,3NF and BCNF with suitable examples.	L4	CO5	**10**